

3(a)	any three from: max two from: 1 (phenotype has) range of values / intermediates ; 2 (phenotypes) not, in groups / in classes / categoric / discrete / qualitative or (phenotypes are) quantitative ; 3 normal distribution ; max two from: 4 polygenic / controlled by many genes ; 5 different genes / alleles, have additive effect ; 6 (named) environmental factors (contribute to the variation) ;	3
3(b)(i)	difference between means = $9.4 - 7.7$ or 1.7 ; denominator = $\sqrt{\frac{0.4^2}{28} + \frac{1.1^2}{27}}$ or $\sqrt{\frac{0.16}{28} + \frac{1.21}{27}}$ or $0.22 / 0.23$ (no limit on number of decimal places) ; $t = 7.56$; must be to 2 decimal places allow ECF	3

Question	Answer	Marks
3(b)(ii)	1 <i>t</i>-value / 7.56, is greater than, 2.01 / critical value ; 2 difference between two groups is, significant / not due to chance or null hypothesis rejected ; any two from: 3 overlap in, data range / error bars ; 4 ref. to correlation does not prove causation ; 5 only a single investigation / needs repeating / small sample size ; 6 ref. to this is only at 5% probability so may not actually be different / AW ; 7 <i>t</i>-test may not be valid due to unequal, standard deviations / variances;	3
3(b)(iii)	any one from: 1 <i>idea of</i> less labour intensive ; e.g. no need for weekly injections / GM only needs to be done once 2 <i>idea of</i> less stressful on GM salmon ; 3 risk of infection with injections ; 4 <i>idea that</i> growth hormone is continuously produced ; 5 GM salmon may grow faster ;	1